WORKED SOLUTION

STEP II 2018, Question 9

Mechanics • Collisions Under Gravity

STEP II 2018, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q9

The key idea

Both beads always have the same acceleration g downwards, so *relative* to each other they move at constant velocity. Once A has bounced, the gap between the beads closes at a steady rate, which gives the time to the collision at once. The last part is then pure energy and momentum: knowing how fast A hits the ground tells us how fast it left the collision, and Newton's experimental law does the rest.

Solution

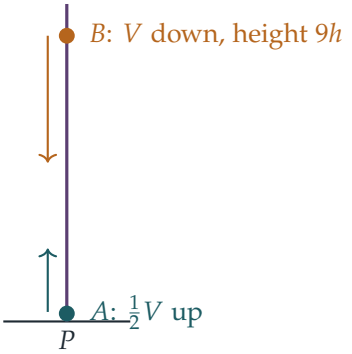
We model the beads as particles of equal mass m on a smooth vertical wire, with no air resistance, so between impacts each bead moves freely under gravity. Take upwards as positive and measure heights from P.

Up to the first bounce

A falls $8h$ from rest, so by conservation of energy

$$V^2 = 2g(8h) = 16gh, \qquad V = 4\sqrt{gh}.$$

In the same time B has also fallen $8h$ from rest (identical motion), so at this instant B is at height $17h - 8h = 9h$, moving downwards with speed V. A rebounds from the ground with speed $\frac{1}{2}V$ upwards.



The collision

From the moment A leaves the ground, both beads accelerate at $-g$, so the velocity of B relative to A stays at its initial value

$$(-V) - \tfrac{1}{2}V = -\tfrac{3}{2}V.$$

Ask yourself Why can we ignore gravity when working out when they meet?

Gravity changes both velocities by the same amount $-gt$, so it cancels in the difference. The gap of $9h$ therefore closes at the constant speed $\frac{3}{2}V$, and the collision happens a time

$$\tau = \frac{9h}{\frac{3}{2}V} = \frac{6h}{V} = \frac{6h}{4\sqrt{gh}} = \frac{3}{2}\sqrt{\frac{h}{g}}$$

after the bounce. (We also need A to be still in the air at that time; we'll see that it is.) A's height then is

$$H = \tfrac{1}{2}V\tau - \tfrac{1}{2}g\tau^2 = 2\sqrt{gh}\cdot\frac{3}{2}\sqrt{\frac{h}{g}} - \frac{1}{2}g\cdot\frac{9h}{4g} = 3h - \frac{9}{8}h = \frac{15}{8}h,$$

as required. Just before the collision,

$$\text{velocity of } A = \tfrac{1}{2}V - g\tau = 2\sqrt{gh} - \tfrac{3}{2}\sqrt{gh} = \tfrac{1}{2}\sqrt{gh},$$
$$\text{velocity of } B = -V - g\tau = -4\sqrt{gh} - \tfrac{3}{2}\sqrt{gh} = -\tfrac{11}{2}\sqrt{gh}.$$

So A is still rising, and

$$u_A = \tfrac{1}{2}\sqrt{gh} \text{ (upwards)}, \qquad u_B = \tfrac{11}{2}\sqrt{gh} \text{ (downwards)}.$$

As a check, B has height $9h - V\tau - \frac{1}{2}g\tau^2 = 9h - 6h - \frac{9}{8}h = \frac{15}{8}h$ too.

The coefficient of restitution

Let e be the coefficient of restitution between the beads, and let v_A, v_B be their velocities (upwards positive) just after the collision. Work in units of $\sqrt{gh}$ for a moment, so $u_A = \frac{1}{2}$ upwards and $u_B = \frac{11}{2}$ downwards.

Momentum (the masses are equal, and the impulse from gravity during the instantaneous collision is negligible):

$$v_A + v_B = \tfrac{1}{2} - \tfrac{11}{2} = -5.$$

Newton's experimental law: the beads approach at speed $\frac{1}{2} + \frac{11}{2} = 6$, and afterwards B, which is above, separates from A at e times this:

$$v_B - v_A = 6e.$$

Subtracting, $v_A = -\frac{1}{2}(5 + 6e)$, so A moves *downwards* with speed $w = \frac{1}{2}(5 + 6e)\sqrt{gh}$.

Now A falls from height $\frac{15}{8}h$ with initial speed w and reaches the ground with speed V (it is below B and moving down faster, since $v_B = -5 - v_A = \frac{1}{2}(6e - 5) > v_A$, so the beads don't meet again first). By energy,

$$V^2 = w^2 + 2g \cdot \tfrac{15}{8}h \quad \Longrightarrow \quad w^2 = 16gh - \tfrac{15}{4}gh = \tfrac{49}{4}gh,$$

so $w = \frac{7}{2}\sqrt{gh}$. Then $\frac{1}{2}(5 + 6e) = \frac{7}{2}$, giving

$$e = \frac{1}{3}.$$

After the collision, A moves down at $\frac{7}{2}\sqrt{gh}$ and B moves down at $\frac{3}{2}\sqrt{gh}$.

Checks

- Kinetic energy: before the collision the total is $\frac{1}{2}m\,gh\,(\frac{1}{4} + \frac{121}{4}) = \frac{61}{4}mgh$; afterwards it is $\frac{1}{2}m\,gh\,(\frac{49}{4} + \frac{9}{4}) = \frac{29}{4}mgh$. Energy is lost, as it must be for $e < 1$.
- With $e = 1$ (and equal masses) the beads would simply swap velocities, so A would leave at $\frac{11}{2}\sqrt{gh}$ and land faster than V; with $e = 0$ they would move together at $\frac{5}{2}\sqrt{gh}$ and A would land slower. The value $\frac{1}{3}$ sits sensibly in between.
- A step-by-step simulation of the whole motion by computer (taking $g = h = 1$) gives a collision at height 1.875 with speeds 0.5 and 5.5, and, with $e = \frac{1}{3}$, a second landing speed of $4 = V$.

Common slips

- Forgetting that B has also been falling while A drops: at the bounce B is at $9h$, not $17h$, and is already moving at speed V.
- Sign errors in Newton's law: write the velocities with one positive direction and check that the separation speed comes out positive.

Beyond STEP

The trick of working in the frame of one falling bead is a first taste of Galilean relativity: in a uniform gravitational field, a freely falling frame sees no gravity at all, which

is the starting point of Einstein's equivalence principle. Collision problems in first-year mechanics are often solved in the centre-of-mass frame, where an impact with coefficient e simply reverses and scales both velocities by e. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both ideas carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.